Name ____ Date ___________ Period _____

Extraneous Solutions and Real-World Applications

Algebra II | Unit 2 | Day 6 | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions when solving absolute value equations, and solve the real-world equations from Day 4.

Vocabulary

Extraneous solution — a value the algebra produces that does ________ actually satisfy the _________________ equation.

Why it happens here — splitting into two cases can create a solution where the "inside" expression has the ___________________ for that case.

Part 1 — A Solution That Does Not Check Out

Solve $3|x - 1| - 2 = 6x + 1$ and check BOTH answers in the original equation.

Isolate: $3|x - 1| =$ _______________ → $|x - 1| = 2x + 1$

Split: $x - 1 = 2x + 1$ or $x - 1 =$ ___________________

Solve: $x =$ ________ or $x =$ ______

Check $x = -2$: $3|-3| - 2 = 7$, $6(-2) + 1 = -11$. $7 \neq -11$ → ____________________

Check $x = 0$: $3|-1| - 2 = 1$, $6(0) + 1 = 1$. $1 = 1$ → __________

Solution set: {______}

Part 2 — Solve Day 4's Real-World Equations

Clarissa's houses: $|x - 1248| = 5$

$x =$ ____________ or $x =$ ____________ Both are even-numbered addresses within 5 houses, so ______________________

Part 3 — Surveying and Brighton Builders

Surveying: $1.5 = |x - 10|/10 \cdot 100$

$|x - 10| =$ __________ $x =$ ____________ or $x =$ __________ miles

Brighton Builders: $2|15x| + 52 = 292$

$|15x| =$ __________ $x =$ ______ or $x =$ ________ (x is a suite count, so use $x = 8$)

That is 8 office suites per side per floor → ________ suites per floor → __________ total suites in the 20-story building

Practice

1. Solve $|x + 3| = 2x$ and check both solutions.

$x =$ ______ or $x =$ ______________________ Solution set: {______}

2. Solve $|2x - 1| = x + 4$ and check both solutions.

$x =$ ______ or $x =$ ________ Solution set: {____________} (both valid this time)

Exit Ticket

Solve $|x - 4| = 3x$, checking both solutions in the ORIGINAL equation. State the solution set.

$x =$ ______ or $x =$ ______________________ Solution set: {______}